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METHODOLOGIES AND APPLICATION

Use of support vector machines, neural networks and genetic algorithms to characterize rubber blends by means of the classification of the carbon black particles used as reinforcing agent

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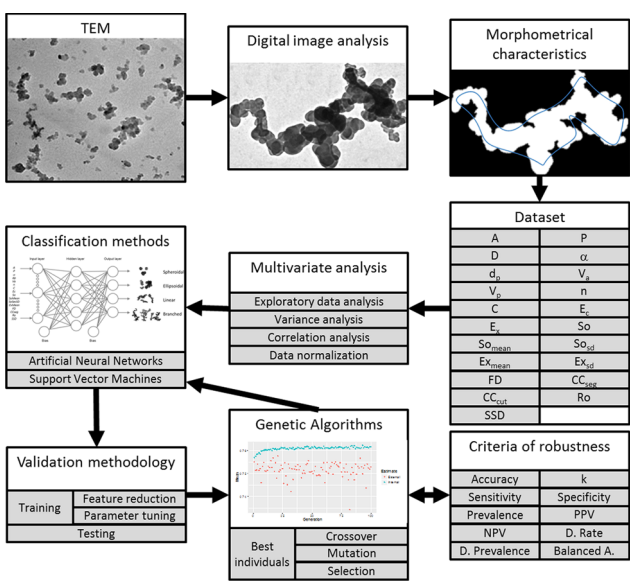
Abstract
In carbon black reinforced rubbers, the shape of the carbon black aggregates has a very significant influence on the final properties of the material. Accurately classifying these particles by shape has proven to be difficult, but the results of the classification would allow to model the final mechanical properties of the material. In this work, 21 features are measured from 7714 isolated filler images obtained from TEM images and used for the classification. Support vector machines and artificial neural network techniques are used to classify the aggregates using a methodology to tune the algorithm parameters to improve the performance of the models. Also, genetic algorithms are applied to make a feature selection in order to get most robust and accurate models. It is demonstrated that the combination of genetic algorithms with support vector machines and artificial neural network improves the classification results and minimizes the complexity of the resulting model.

Graphical abstract



Keywords Reinforced rubber blends · Digital image analysis · Aggregate shape classification · Artificial neural networks · Support vector machines · Genetic algorithms

1 Introduction

Rubber blends are often reinforced using carbon black (CB), which consists of carbon-based spherical nanoparticles fused

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together to form aggregates, and those aggregates forming agglomerates of different shapes. The degree of reinforcement is defined by several parameters, like the filler shape, size, aspect ratio, and filler–matrix interactions (Goettler and Cole 2001; Mark et al. 2005). The shape of the particles has a significant influence on the microstructure of the rubber compound, and consequently on its electrical and mechanical properties (Hess et al. 1969, 1973).

These CB shapes are usually classified into four categories (Herd et al. 1992; Lopez-De-Uralde et al. 2012), which correlate with the properties of composites. These four morphological categories are: spheroidal, ellipsoidal, linear, and branched (Fig. 1).

In order to obtain automatically the shape of the reinforcement fillers, many techniques have been applied to images obtained from transmission electron microscopy (TEM). These techniques are basically applied to define several aggregate features from an automated image analysis (Luengo et al. 2015; Maas and Gronski 1999). Once the main features of the aggregates are calculated, soft computing and machine learning techniques, which have proved to be accurate and efficient methods to solve complex problems (Fernandez Martinez et al. 2011; Lostado et al. 2016), can be applied to classify the aggregates by shape (Fernandez Martinez et al. 2017).

In this work, support vector machines (SVM) and artificial neural networks (ANN) combined with genetic algorithms (GA) are used to classify nanoparticle aggregates of carbon black into one of the morphological classes that were previously proposed in the literature. In the proposed approach, TEM image processing is performed to measure several morphological characteristics of each aggregate. Then, a wrapper feature selection algorithm based on GA is applied to select the most significant features to optimize the classification models. In this case, each individual of each generation is a binary subset of features, where a feature is either included or not according to the binary value. At the same time, each individual is used to obtain the validation of several classification models obtained from SVM and ANN algorithms while fully tuning of the most significant parameters of each technique. Also, and to demonstrate the efficiency of the method, the resulting models with and without GA optimization will be studied and compared.

The selection of the most accurate model among the built ones is performed using an objective function evaluated for each of the individuals. That function is based on several robustness criteria commonly used in classification problems, like the overall classification accuracy and the unweighted kappa statistic. The final goal of this work is to select the most accurate model to automatically classify samples of aggregates extracted from any CB grade directly from TEM images, detecting which are the most significant features that define the problem, and studying the classification

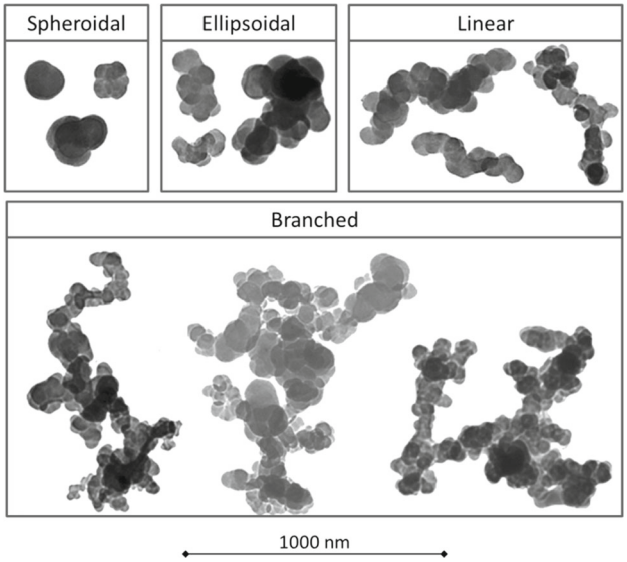

Fig. 1 Morphological categories of carbon black aggregates

robustness criteria to understand the behavior of the selected model for each of the classes. The results of the classification are of interest to find the relationships between the processing parameters, the resulting microstructure and the final properties of CB reinforced rubbers.

This study is the culmination of several works performed during the last years, where a methodology to classify carbon black aggregates used as reinforced particles on rubber blends was taken and improved. Both the methodology and the results have been improved in several works. First, Fernandez Martinez et al. 2014 developed some classification models based on a small group of particles (48 samples of a particular grade, Vulcan XC605) that were selected because their shapes could be indistinctly classified into the four considered classes. Despite the high accuracy results obtained in this work, when the built models were validated with new samples, it was concluded that the boundaries that limits the classes on the classification models were overtrained, and the error resulting for the classification of the larger datasets was significantly larger. Thus, new models, in this case trained with a bigger dataset of aggregates, should be created. So, Fernandez Martinez et al. (2017) generated a new database with a higher number of samples (7714 of several carbon black grades: Vulcan XC 605, Vulcan XC 72, CSX 691, Printex 25, N990 and N762). Then, the previously used methodology was implemented on this extended dataset, additionally using several new machine learning techniques, resulting in a better generalization, higher accuracy, as shown by the better classification results when the models were validated with new instances. Neither of these two works performed a complex feature selection from the original set of variables, although some techniques like principal component analysis (PCA) or analysis of variance were applied to study, transform and reduce the number of the original

Table 1 Dataset used: number of TEM images, number of isolated aggregate images, and aggregate classification

Grade of CB	Number of TEM images	Number of aggregates	Class of each aggregate			
			Spheroidal	Ellipsoidal	Linear	Branched
Vulcan XC 605, Vulcan XC 72, and CSX 691	62	2596	121	1240	446	789
Printex 25	61	3117	232	2004	402	479
N762, and N990	19	2001	264	1244	116	377

features. Thus, the most significant contribution of the current work is the use of a complex feature selector, based on evolutionary techniques like genetic algorithms, that allows studying the influence of each original feature in the classes to subsequently reduce the number of original features without any transformation that will be part of the classification model. In this way, determine if the accuracy of the models can be improved reducing the number of features.

2 Materials

Several CB grades (Table 1) have been analyzed in order to have a significant and balanced dataset of samples, which should result in more robust classification models. Vulcan XC 605, Vulcan XC 72, and CSX 691 are high structure special grades where more than 30% of the aggregates are classified as branched. Printex 25 is also a special grade where more than 15% are classified as branched as well. Finally, N990 and N762 are grades that are formed by larger particles and lower structure, in which around 75% of the aggregates are classified as spheroidal or ellipsoidal.

3 Methods

3.1 Morphological characterization

The CB samples were prepared according to the ASTM standard D3849-07 (ASTM Standard D3849-07 2011). The CB powder was dispersed in a liquid (chloroform) and deposited on a TEM grid and let dry. Then, several TEM images were acquired for each powder and automatically processed following the methodology proposed by Fernandez Martinez et al. (2014), isolating each of the aggregates and measuring twenty one parameters for each of them.

The twenty-one parameters under study were defined based on three different approaches: Several of the used parameters are described in the ASTM Standard D3849-07, a second group of parameters are on the boundary of the aggregates, and finally there are those based on the region of the aggregates. The measured features based on the ASTM Standard D3849-07 are: aggregate area (A), and aggregate perimeter (P), diameter of the equivalent circle (D), aggregation factor (α), average particle size for the aggregate (d_p), aggregate volume (V_A), particle volume (V_p), and number of particles in the aggregate (n). The measured features based on the boundary of the aggregates are: roughness (Ro), signature (SSD), number of Fourier descriptors that define the shape (FD), number of random segments that intercept the boundary ($CCseg$) and total number of interceptions ($CCcu$). The parameters based on the region of the aggregates are: circularity (C), solidity (S_o), extent (E_x), eccentricity (E_c), mean and standard deviation of extent (Ex_{Mean}, Ex_{SD}), and mean and standard deviation of the solidity (So_{Mean}, So_{SD}). A detailed explanation of these 21 features can be found in the work done by Fernandez Martinez et al. (2014).

In addition to the computed 21 features, all 7714 aggregates were visually classified according to their shape by five (experienced) members of our research group in order to assign the class to each aggregate. Once the final dataset with the values of the 21 defined features for each of the 7714 aggregate of nanoreinforcement fillers was calculated and stored, a multivariate analysis was performed concluding that all the features analyzed versus the classes are significant.

3.2 Classification method

The proposed approach uses the combination of GA and two classical but powerful classification techniques—i.e., ANN and SVM—to classify the aggregates by their shape. These two techniques are compared as well with classical classification trees (a simpler method having a lower computational cost), to analyze the convenience of their use.

3.2.1 Supervised feature selection using genetic algorithms

Genetic algorithms (GA) are stochastic search algorithms inspired by the basic principles of biological evolution and natural selection (Holland 1975; Mitchell 1998). GA have been successfully applied to solve many optimization problems, both for continuous and discrete functions (Lostado et al. 2015; Sanz-Garcia et al. 2012).

In this case, GA have been applied to make a selection of the most significant variables or features. The problem of feature selection is inherently a complex optimization problem, where the combination of features that provides an optimal prediction of the response is sought (Wan et al. 2016). This process conducts a search of the feature space

where the chromosome or individual is a binary vector that has the same length as the number of features in the dataset (Kuhn and Johnson 2013; Scrucca 2012), see Fig. 2. The search process is initiated with a random selection of chromosomes from the population of all possible chromosomes. Then, an objective function is defined for each algorithm that classifies the aggregates, in this case ANN or SVM. Afterward and according to this function, the likelihood of each chromosome is determined and used to obtain new chromosomes. Finally, selection, crossover, and mutation operators are applied to create the next generation of chromosomes in order to evolve toward better solutions, granting a higher chance of being reproduced to the chromosomes with higher fitness values. This stage was performed using the R statistical language environment (Kuhn 2015; R Core Team 2016).

The training data are split following a tenfold cross-validation scheme. For each fold, nine-tenths of the data are used in the search through the estimation of an internal performance, while the remaining tenth is used to estimate the more significant external performance since these instances are not used in the training. Both results are taken into consideration for the study: the external and the internal performance. Then, the entire dataset is used in the last execution of the GA search and the final model is built with the subset associated with the optimal generation determined by resampling.

This entire process was repeated 20 times since some methods, like ANNs, use random initializations (random initial values in their learning procedures). Therefore, the process has to be repeated several times to improve the statistical significance of the obtained results.

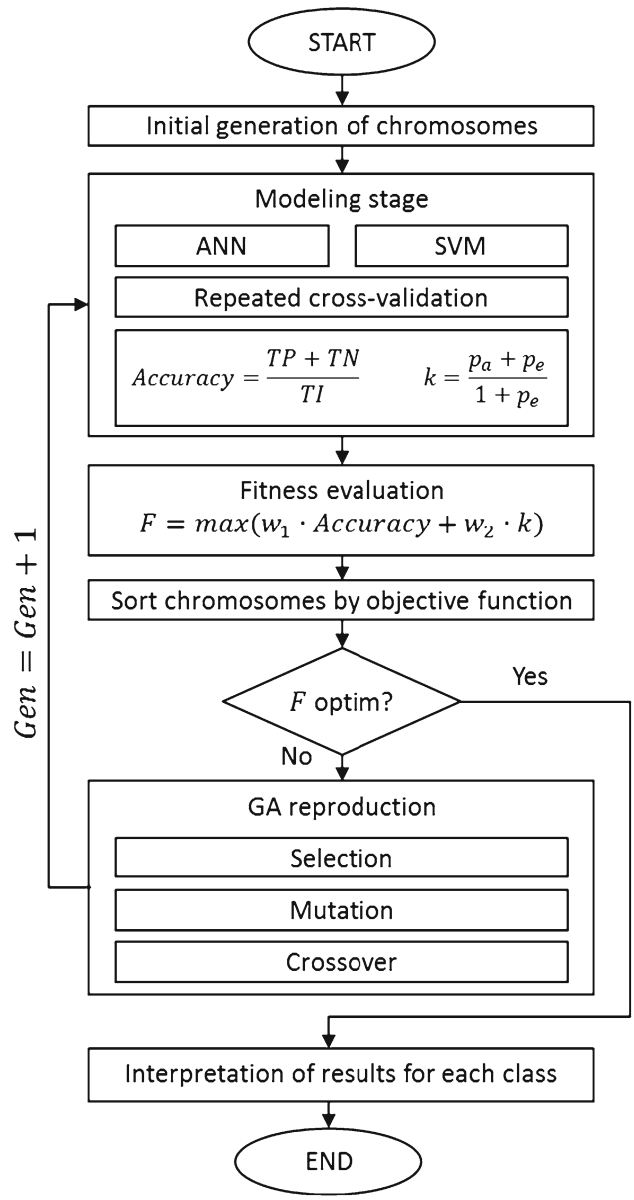

Fig. 2 Flowchart of the GA + (ANN-SVM) process

3.2.2 Support vector machines

SVM is a supervised learning method that generates input–output mapping functions from a set of labeled instances (Cortes and Vapnik 1995; Vapnik 1998; Weston and Watkins 1999). When the aim is to make a classification, this mapping function is used to categorize the input dataset. This classification method is based on nonlinear kernel functions that are used to perform a transformation of the initial dataset to a higher-dimensional feature space, where the ideal boundaries to split classes can be found.

The initial SVM studies focused on a two-class dataset but soon a two-class classification was conducted to a multiclass classification, and the most used solutions in these cases are one-against-one (OAO) and one-against-all (OAA) SVM (Vapnik 1995). These two SVM methods convert a multiclass classification problem into n two-class problems reducing a multiclass problem to binary ones. In this work, the OAO approach is used since it has been reported to perform better (Hsu and Lin 2002).

In OAO-SVM, the decision functions for all the combinations of class pairs is determined. In this method, $\frac{k(k-1)}{2}$ binary classifiers are trained, and Eq. (1) is solved for each binary problem (ith class and jth class).

$$\begin{aligned}
&\min_{\omega^{ij}, b^{ij}, \xi^{ij}} \frac{1}{2}(\omega^{ij})^T \omega^{ij} + C \sum_t \xi_t^{ij} (\omega^{ij})^T \\
&(\omega^{ij})^T \phi(x_t) + b^{ij} \geq 1 - \xi_t^{ij}, if\ y_t = i \\
&(\omega^{ij})^T \phi(x_t) + b^{ij} \leq -1 + \xi_t^{ij}, if\ y_t = j \\
&\xi_t^{ij} \geq 0
\end{aligned} \tag{1}$$

Table 2 Classification techniques applied in the analysis, with a brief definition of the parameters tuned during the training stage, and their selected values

Classification technique	Parameters	Range	Optimal value
ANN	**size**: number of neurons in the hidden layer	1: 20	5
	decay: parameter to decrease the learning rate of the optimization function	0.001:0.1	0.01
SVM-L	**C**: cost of constraints violation	0.2:10	0.6
SVM-P	**C**: cost of constraints violation	0.2:10	1.5
	degree: degree of the polynomial kernel function	2:3	2
SVM-RBF	**C**: cost of constraints violation	0.2:10	0.8
	sigma: inverse kernel width used by the Gaussian kernel	0.1:1	0.2
CART	**cp** (complexity parameter): Any split that does not decrease the overall lack of fit by a factor of cp is not attempted	0:2	0.01

Table 3 Selection of the most significant features according to the GA process

	CART	ANN	SVM-L	SVM-P	SVM-RBF
A	x	x	x	x	x
P	x	x	x	x	x
D	x		X	x	x
α		x	x	x	x
dp	x	x	x	x	x
Va	x	x	x		
Vp			x	x	x
n	x	x			x
C	X	x	x	x	X
Ec	X	X	X	X	X
Ex	x		x	x	x
So	X	X	X	X	X
SoMean	x	X	x	x	x
SoSecSD	x	x	x	x	x
ExMean	x	x	x	x	x
ExSD	x				
FD	X	X	x	X	X
CCseg		x			x
CCcut	x				x
Ro	X	X	X	X	X
SSD	x	x	X	X	x

The top five selected features in each case are in upper case x (X)

When all the classifiers are built, a decision considering all of them must be taken. In this case, a voting strategy is used by which the most voted class is chosen. When $sign((\omega^{ij})^T \phi(x) + b^{ij})$ indicates x is in the ith class, the number of votes for ith class is increased by one. The same operation is performed for the jth class as well.

In this method, every data point is mapped using $\phi(x)$ as mapping function into a higher-dimensional space via some transformation using three kernel functions: linear (SVM-L), polynomial (SVM-P) and Gaussian radial basis function (SVM-RBF).

3.2.3 Artificial neural networks

Neural networks are among the classification techniques having nonlinear nature one of the most intensively studied and applied (Bishop 1995; Ripley 1996). This technique is inspired by actual brain neural networks. The ANN usually present a system of interconnected neurons with numeric weights and biases that can predict the value of a class.

Many approaches on how a neural network can classify a class from several inputs have been studied. In this work only the classical feed-forward neural network defined by Ripley (1996) is used, a one-hidden-layer network where the model is given by Eq. (2).

$$y_u = g(x_i) + e_i = \sum_{k=1}^{s} \omega_k g_k \left(b_k + \sum_{j=1}^{p} x_{ij} \beta_j^{[k]} \right) + e_i; i = 1, \ldots, n \quad (2)$$

where $e_i \sim N(0, \sigma_e^2)$, s is the number of neurons, ω_k is the weight of the each neuron, b_k is the bias for each neuron, $\beta_j^{[k]}$ is the weight of each input of the network, and $g_k(.)$ is the hyperbolic tangent activation function $g_k(x) = \frac{e^{2x}-1}{e^{2x}+1}$.

The mean-squared error of the neural units is calculated and the parameters (weights and bias) optimized to minimize the error. This optimization is based on a gradient descent with momentum weight function that is used as an adaptation learning function (Bishop 2006).

3.3 Robustness criteria

The robustness of the models is a commonly accepted parameter that allows selecting the optimum model. In this case, this robustness is assessed by using the overall classifica-

Table 4 Results obtained during the classification stage, both using and not using GA

	With GA			Without GA	
	Accuracy (%)	Kappa	Time (h)	Accuracy (%)	Kappa
CART	70.94	0.50	7.57	69.80	0.48
ANN	75.75	0.58	210.06	74.80	0.56
SVM-L	74.94	0.55	609.33	74.90	0.58
SVM-P	74.25	0.55	1757.43	74.10	0.54
SVM-RBF	74.07	0.53	117.42	74.60	0.54

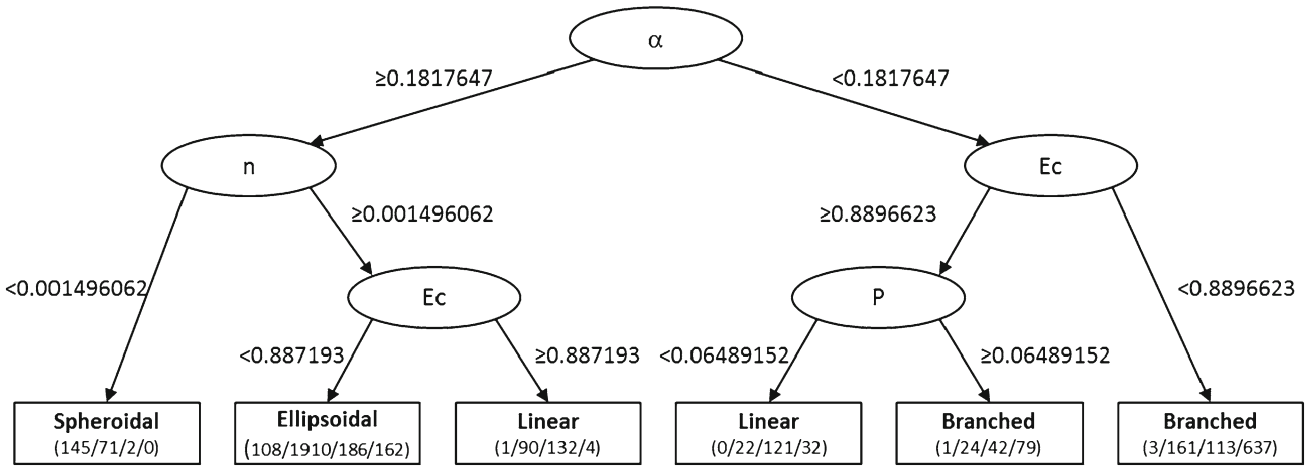

Fig. 3 CART tree obtained after using GA as feature selector. Numbers in brackets show: number of instances classified like Spheroidal/Ellipsoidal/Linear/Branched

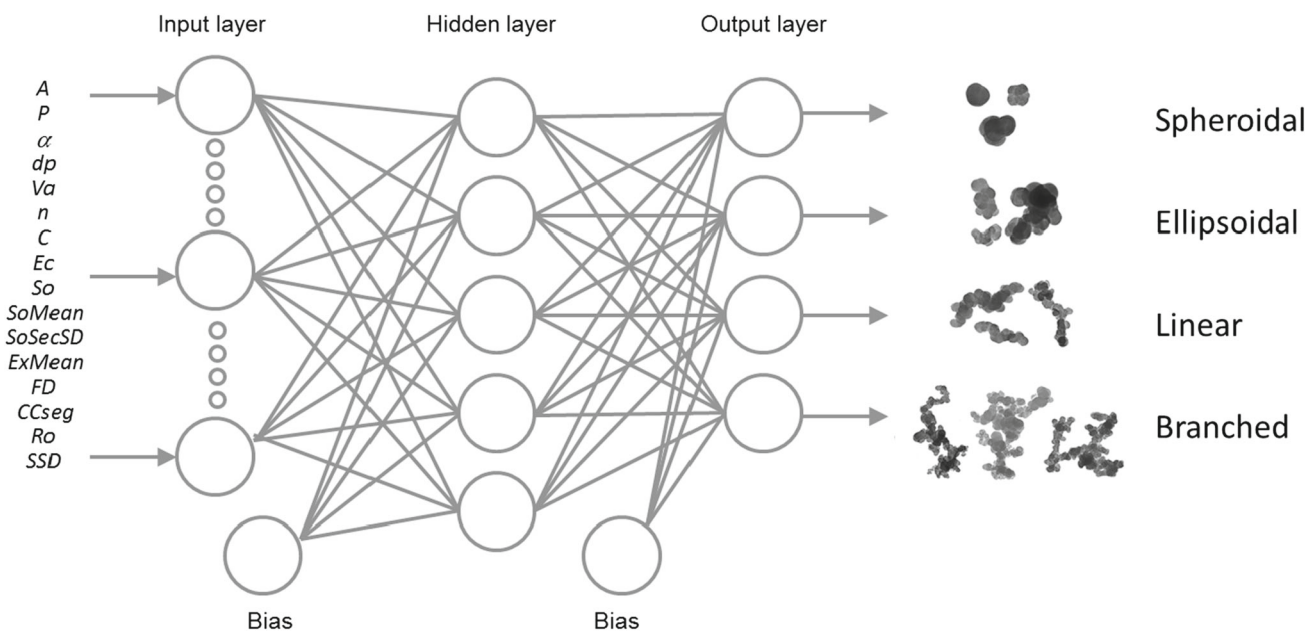

Fig. 4 Structure of the ANN obtained after using GA as feature selector

tion accuracy [Eq. (3)] and the un-weighted kappa statistic [Eq. (4)].

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TI}} \tag{3}$$

$$k = \frac{p_a + p_e}{1 + p_e} \tag{4}$$

where TI is the number of total instances, TP is the number of aggregates that have the condition and test positive for

Table 5 Weights of the neurons on the hidden layer of the ANN

	Neuron 1	Neuron 2	Neuron 3	Neuron 4	Neuron 5
Bias	– 1.61535607	– 0.19891104	– 5.62142480	– 1.24594549	– 1.84813570
A	– 1.74872647	4.72708099	– 0.93373084	2.23226439	7.53297628
P	– 1.92648098	16.19711309	1.52541919	3.80351336	4.95630149
α	– 11.36137630	0.85579704	– 5.18481921	7.85436626	7.49031517
dp	– 3.20630442	2.17448473	0.48125917	8.56702445	5.50299536
Va	– 0.83771332	0.27944163	– 3.01332531	– 5.03405747	8.42980116
n	– 0.80831005	2.77477303	7.81058988	0.02448841	– 1.29258819
C	– 8.23046219	2.76908960	12.64832815	3.74765887	– 6.39894119
Ec	– 2.61798934	– 2.61002670	2.08496783	1.30355785	– 4.46361988
So	– 0.03994204	0.80924317	– 2.54274534	1.74614578	– 1.07746974
SoMean	3.23812947	– 0.94753625	1.11024414	– 4.49387720	1.84411060
SoSecSD	– 2.67906370	– 0.13992430	0.91083801	– 1.89011817	1.64050954
ExMean	– 4.36024669	– 0.67338004	0.90857724	– 0.70273015	0.24028047
FD	11.13000574	– 2.18031729	– 8.69027384	– 2.86772904	0.69789766
CCseg	1.45322199	0.56907380	0.96001245	– 2.63156987	5.90865709
Ro	7.38334328	6.28541980	– 4.43649974	3.82083010	9.17796980
SSD	– 3.66033947	– 7.77643462	0.53113957	0.59591262	– 9.67495186

the condition, TN is the number of aggregate that do not have the condition and test negative for the condition, p_a is the proportion of observations in agreement, and p_e is the proportion in agreement due to chance.

4 Results and discussion

The generation of classification models that classify the shape of nanoreinforcement fillers was based on the combination of GA and classical machine learning techniques based on ANN and SVM. Also, and to have the possibility to compare high and low computational cost techniques, classical classification and regression trees (CART) were applied too.

Firstly, the 21 defined features were measured for 7714 isolated CB images obtained from TEM images of the dispersed CB powder. Previously to the use of classification methods, all the variables in the dataset were normalized between 0 and 1 to balance their weight and improve the quality of the statistical analysis.

Then, these 7714 instances were used to apply the method GA + (ANN-SVM) and build the classification models using cross-validation. During the process, the chromosomes that define each individual of the generation are modified based on several random decisions, for example, number of bits from each parent that will end up in the crossover. This operation can produce a variability of the results defined by the accuracy, and to solve this problem the procedure was repeated 20 times and the mean values were calculated to define the final accuracy criterion. The overall classification accuracy was computed and used as performance indicator of the models. Simultaneously, during this training, a tuning of the most important parameters of each algorithm was performed to improve the prediction capability. The parameters that were tuned in the analysis, the range of tuning, and the selected optimal values to build the final models are shown in Table 2.

If the complexity of the model—commonly defined as the number of input features and the internal structure of the model—is high, it is subject to overfitting and generalization will be poorer. Therefore, it is desirable to keep its complexity low. GA were used to select the most significant variables in each case (Table 3), simplifying the model. During the GA operation, the maximum number of generations was set to 100, the number of individuals per generation to 50, the percentage of individuals obtained from the elitism process was 10%, another 10% was obtained from the mutation process, and the final 80% was obtained from crossover between individuals.

The classification was accomplished by using the selected features in each case. Table 4 shows the overall classification accuracy and the un-weighted kappa statistic for the proposed algorithms, both using and not using GA to simplify the models. In most of the cases the accuracy was improved using GA, and moreover the improved performance was achieved incorporating fewer features into the models. For example, the best ANN model showed a 75.75% overall classification accuracy, improving the 74.80% accuracy obtained without GA optimization and reducing the number of features from 21 to 16.

Clearly, the methods that are more computationally intensive (as compared to CART) result in more accurate models. However, CART has the advantage of providing the model

Table 6 Weights of the neurons on the output layer of the ANN

	Spheroidal	Ellipsoidal	Linear	Branched
Bias	– 13.85702300	1.69919681	5.70532342	6.45057105
Neuron 1	– 1.02130411	– 3.14124820	– 0.32473081	4.53545641
Neuron 2	15.94546999	– 0.32954336	– 15.16490274	– 0.52938217
Neuron 3	6.72075402	– 0.26860676	1.58985865	– 7.97547377
Neuron 4	– 0.02003883	1.94485398	0.44239952	– 2.29451344
Neuron 5	– 2.96615052	– 1.73844353	7.58903669	– 2.78622859

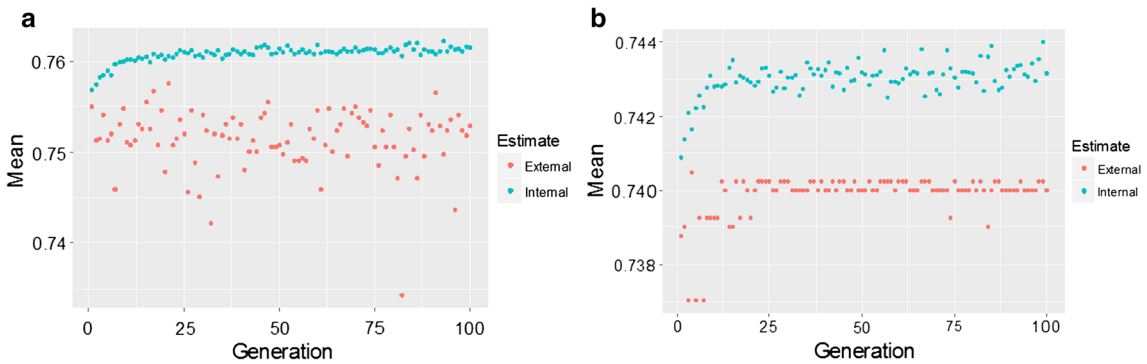

Fig. 5 Comparison of the GA process between **a** GA + ANN and **b** GA + CART

in a form that is easier interpret and understand, as shown in Fig. 3. The tree of the figure shows that each instance that ended up in a given leaf was classified as belonging to the class of the leaf, even if the instance belonged to another class. For example, of the 218 instances that reached the leaf classified as spheroidal (and therefore were classified as being spheroidal), only 145 instances were actually spheroidal, 74 were ellipsoidal and 2 were linear.

Analyzing the results obtained from all the models it can be concluded that ANN results in the most accurate classification. However, the computational cost of the ANN-based method compared to that of CART is close to 30 times larger. Still, the computational cost of the generation of ANN-based models is slightly lower than those for SVM-L and SVM-P. Considering that the models were only calculated during this stage and thereafter they can be used as often as desired, the final model chosen (Fig. 4) was an ANN with 5 neurons in the hidden layer and with a final classification accuracy of 75.75%. Tables 5 and 6 show the weights of the elements that set up the selected ANN.

Finally, Fig. 5 shows the results of the whole GA process when applied to the ANN and CART techniques. The mean (internal and external) accuracies for each of the 100 generations are shown in the figure. Moreover, it is shown how the internal performance accuracy is higher than the external performance, but this is an expected consequence of an overtraining during the resampling and the cross-validation training. Therefore, the external performance was chosen to study the robustness of the models.

5 Conclusions

In this work, a methodology that applies data mining processes based on soft computing and machine learning techniques has been applied and validated. Its aim is to create models that can automatically classify the particles in any carbon black sample according to their morphology, based on TEM images. Also, the effectiveness of the use of GA as feature selector for minimizing the complexity of the models has been assessed and verified.

The proposed method shows that the most accurate model was obtained using GA + ANN. Moreover, considering the better results in comparison with those obtained using ANN without GA and also using other techniques like SVM and CART, it was concluded that the methodology using GA + ANN was successful in characterizing the carbon black filler by classifying its aggregates into specific morphological categories. Besides the improvement in the accuracy, the feature reduction applied resulted in a model that reduced the number of variables minimizing the complexity of the model. The proposed methodology has the drawback of involving a significant increase in computational cost, but the benefits in terms of improvement in the accuracy—in a highly complex, nonlinear problem—outweigh the higher computational cost.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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